

FIFTH BIENNIAL REPORT

OF THE

STATE VETERINARIAN

OF CALIFORNIA

FOR THE TWO YEARS ENDING JUNE 30, 1910

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REPORT OF THE STATE VETERINARIAN.

To His Excellency, JAMES N. GILLET, Governor of California.

In presenting the biennial report of this department for the sixtieth and sixty-first fiscal years, it is very gratifying for the State Veterinarian to report the continued progress that has been made in the work of controlling and eradicating some of the communicable diseases of live stock within the borders of this State, and the prevention of disease being carried in from other states and territories.

It is also most pleasing to report the advance in prosperity of our live stock industry. It might be well in this report to mention some facts in reference to the live stock industry in California, showing the enormous amount of capital invested therein and what its protection means to this State.

On January 1, 1910, the number, average price, and farm value of horses, mules, sheep, milch cows, and other classes of cattle and hogs in California are given in the report of the Secretary of the United States Department of Agriculture, as follows:

	Number.	Average price per head.	Farm value.
Horses -----	420,000	\$105 00	\$44,100,000 00
Mules -----	83,000	122 00	10,126,000 00
Sheep -----	2,372,000	3 30	7,827,600 00
Milch cows -----	452,000	38 40	17,356,800 00
Other cattle -----	1,120,000	20 10	22,512,000 00
Hogs -----	540,000	8 20	4,428,000 00
Totals -----	4,987,000	-----	\$106,350,400 00

It can be readily seen from the above figures that our live stock industry is an enormous one and should have adequate protection. Irregardless of new innovations, such as gas engines, electrical inventions and automobiles, the horse and mule industry of this State has prospered; in fact, during the past decade horses and mules have more than doubled in value, and have increased very materially in numbers.

The value of all classes of cattle has also shown marked advances during the past decade. While there has been a slight decrease in the number of range cattle there has been a large increase in the number of milch cows. This is due to the fact that many of the large ranges

in the Sacramento and San Joaquin valleys are constantly being cut up into smaller ranches, and with the development of irrigation are turned into dairies. The dairy industry is constantly growing in California, and, with our ideal conditions for such an industry, it is only a question of time when California will be one of the leading dairy states of the Union.

With the rapid development of our dairy industry there should have been a corresponding increase in our swine production. This industry has, however, not made any appreciable strides in point of numbers during the past few years. The reason for this is due solely to the fact that the raising of hogs is an hazardous business on account of prevalent communicable diseases, such as cholera. This subject will be more extensively dwelt upon later on in this report, and I shall endeavor to show how our swine industry can be encouraged and built up by adequate State protection.

By the enactment of suitable laws in the last session of the legislature, this department has been able to extend its sphere of usefulness to the live stock industry. The act requiring all veterinarians in the State to immediately report outbreaks of infectious diseases to the State Veterinarian has uncovered many foci of disease which hitherto would never have come under our observation. This is particularly true in the case of the disease known as glanders, in horses and mules. Upon receiving a report of such a disease, this department immediately proceeds to investigate and uncover, if possible, any other cases in the immediate neighborhood of the affected animal. All contact horses and mules are tested with mallein. Furthermore, all affected animals are now destroyed in conformity with section 402*b* of the Penal Code, and proper sanitary precautions are taken to prevent further spread of the disease.

The work of this department not only deals with the live stock industry, but indirectly it has an important relation to human health. A number of animal infections are directly communicable to man, and in our work in handling such diseases it is essential that precautions be taken to guard against the spread of the infection to man. This is particularly true in such diseases as anthrax, glanders, and rabies. In several instances these animal diseases have caused fatalities in human beings in California during the past two years.

During the two years just past this department has paid particular attention to the suppression of such infections as glanders in horses and mules, anthrax in all animals, the cattle tick, scabies in sheep, and rabies. While we have endeavored to investigate every reported outbreak of disease, we have found that by centering our efforts in the suppression and eradication of certain diseases known to exist in various parts of the State, better results could be accomplished than

by scattering our efforts. This is particularly true in the fight against the cattle tick (*Margaropus annulatus*), scabies in sheep, and glanders in horses and mules.

CATTLE TICK ERADICATION.

Our policy in continuing to systematically eradicate the cattle tick (*Margaropus annulatus*), the carrier of Texas or Southern cattle fever infection in cattle, has resulted in the freeing of many herds of cattle from this parasite during the past two years, and the consequent removal from State and Federal quarantines of large areas of territory. In this work California has had the coöperation of the United States Bureau of Animal Industry, working under a special appropriation from congress to coöperate with quarantined states. The Federal, State, and County live stock inspectors supervise the disinfection of cattle and maintain State quarantines in conformity with State laws on the subject, and also make inspections of cattle in the provisionally quarantined areas for movement to noninfected areas in this State.

Every infested herd is disinfected at regular intervals by the inspector of the district. The parasiticide used is an arsenical solution that is first mixed by the inspector, in order that the solution may be of proper potency to kill the ticks but not injure the cattle. Infested cattle are dipped every thirty days during the tick season, with a view of destroying the ticks before the females have matured sufficiently to drop off and ovulate. In this manner regeneration of the tick is prevented, and under ideal conditions all ticks on a ranch are destroyed in one year. However, ideal conditions are not always obtained in a work of this magnitude, and from a number of causes this work of cleaning a ranch may be prolonged several years.

Ticks have been eradicated from nearly all of our larger cattle ranges by the simple method of removing all the cattle, horses and mules from the infected range for a period of at least eight months, after which time the parasites die from starvation. During the past two years, however, tick infestation has been confined to the smaller beef ranges and dairies, in the majority of which places tick eradication has been conducted through means of dipping in the arsenical solution on account of the fact that through lack of sufficient range it is impossible to carry out the starvation method.

Our work in this line has met with continued success. In fact, California is far in the lead of other tick infested states in results accomplished. During the past two years we have kept pace with the work accomplished in the two years before, as is evidenced by the following area of territory freed from ticks and taken out of quarantine.

During 1909 the territory removed from the Southern cattle fever quarantine was as follows:

All of Imperial County.

All of those parts of San Bernardino and Riverside counties situated east of the one hundred and sixteenth meridian, west longitude.

San Diego and Orange counties were placed in the provisionally quarantined areas, from which cattle can be moved at all times of the year after they have been inspected and found free from tick infestation. This privilege was granted to these two counties on account of the fact that the work has progressed so well therein that only a comparatively small number of herds were still tick infested, and these being under strict quarantine and undergoing process of cleaning, there was no danger to outside cattle in allowing cattle from these counties to move after they had been inspected and found tick free.

On March 1, 1910, by proclamation issued by your Excellency, there was removed from all quarantine restriction and placed in the free area the following territory: All of Los Angeles and Ventura counties; the balance of Riverside County, *i. e.*, that part of Riverside County situated west of the one hundred and sixteenth meridian, west longitude; and the balance of San Bernardino County, with the exception of six townships situated near the Mojave River.

There was also removed from quarantine, in same proclamation, all that part of San Luis Obispo County situated east of the summit of the Santa Lucia range of mountains.

This leaves at the present time the following areas in this State quarantined for cattle ticks: San Diego, Orange, and Santa Barbara counties; the western part of San Luis Obispo County; one township in Fresno County; one township in Kings County; two townships in Tulare County; and six townships in San Bernardino County. These districts are only provisionally quarantined, however, and cattle can be moved from them to other sections in this State, after the cattle have been inspected and found tick free.

While the coöperation we have received from the stockmen in this work has been practically unanimous, we have had in a number of instances some strenuous opposition, and in some of these instances it has been found necessary to resort to the forcible seizure and disinfection of cattle under the provisions of the act of the legislature that gives the State Veterinarian authority to seize and disinfect a herd of tick infested cattle, and hold same as a lien for the expenses incurred in so doing, in case the owner refuses to disinfect his cattle. In other cases the mere threat to use this power has been sufficient to force owners of infested cattle to dip.

With the rapid cleaning up of the tick infested area of this State, we have the gradual elimination of the immune cattle. The cattle now raised on these tick free ranches of the south are just as susceptible to Southern cattle fever as are the cattle of the north. This fact necessitates very careful observance in the maintenance of quarantines. I

am pleased to report, however, that only in a few instances were our quarantine regulations infringed. One party was convicted and fined in Kings County, while another was fined in San Diego County, and a third party was acquitted in San Diego County on technical grounds.

While it is only a matter of a few years ago that Southern cattle fever was the bane of California cattlemen in the central and southern counties, resulting in great losses of cattle annually, I am pleased to report that during the past two years the fatalities in California cattle from this disease were practically nil.

A number of reports have been received in this office during the past two years of cattle infested with fever ticks in counties located outside of the quarantined area. Each report has been investigated through this department, but no Texas fever ticks (*Margaropus annulatus*) were found. In each instance other varieties of harmless ticks were found infesting cattle, their identity having been confounded by those reporting with the fever ticks.

During the past two years 463 inspections, aggregating 29,519 head of cattle, were made for movements of cattle from the quarantined area to points outside of same in this State.

GLANDERS.

This disease continues to be a menace to the horse and mule industry of this State. Since the enactment of the act of March 19, 1909, requiring the reporting by veterinarians of glanders to this office, more or less systematic efforts have been made to eradicate this infection. However, some conception of the prevalence of this disease among equines in California might be gleaned from the number of reports that we have received after the passage of this act.

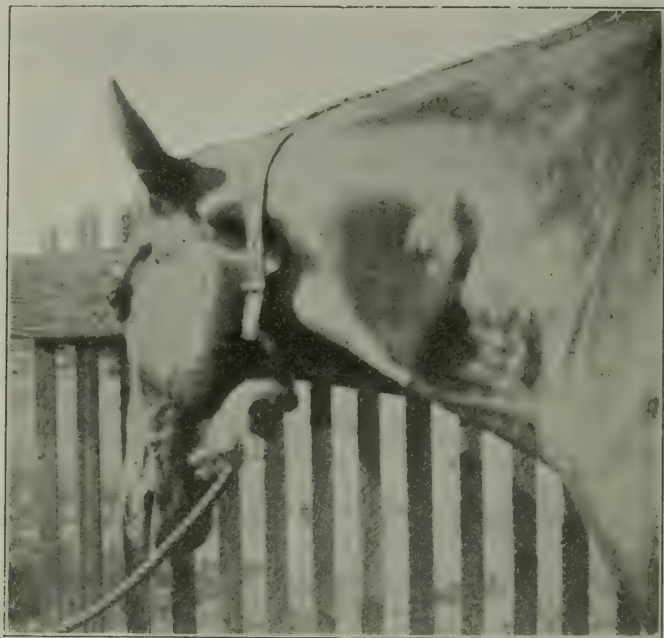
Prior to the enactment of this law a horse or mule affected with glanders might be brought to a veterinarian, who, after diagnosing the disease, would advise the owner to destroy the animal and test the contacts. While in some instances the veterinarian's advice would be followed, in other instances, however, it would not. In the latter case it was more than likely that the affected animal would be sold or traded, and in this manner scatter the disease to other animals. It was a very common practice to hear of a horse affected with chronic glanders change ownership a number of times. Just as soon as the owner suspected or was informed of the condition of the animal, he traded it off or sold it. Of course in a number of instances this was done without the owner having a true knowledge of the dangerous nature of the disease. But the result was just the same in the infecting of other animals.

With the enactment of the law requiring the immediate reporting of such cases to the State Veterinarian, we have been enabled to

early investigate each outbreak of this disease, cause the clinically affected animals to be destroyed, and test the contact animals for the purpose of ascertaining if any of them are affected internally without exhibiting external symptoms of this disease.

It can be readily seen how necessary it is for this department to obtain an early report of an outbreak of glanders, in order that adequate measures may be taken to exterminate the disease and prevent its further dissemination among healthy animals.

Quite as important is the detection of occult cases of this disease,



Case of occult glanders, showing well marked local reaction to mallein.

i. e., those cases in which the lesions are confined to the internal organs. It is here that the difficult part of our work is presented. After testing a number of animals with which one clinical case of glanders has been in contact, we invariably find one or more animals which react to the test, showing that they are affected with the disease internally. In many instances these animals appear to be, from all external appearances, just as healthy as animals which have passed the test, and it is a difficult matter at times to convince the owners that these animals are also affected with glanders.

Glanders is similar to tuberculosis in many respects, one of which is that an animal may be affected and still show very little loss of condition. Then, again, infection in these animals may be of recent

origin and the lesions so slight as not to affect as yet the general economy of the animal. Still if these animals are allowed to live and constantly come into contact with healthy animals, they are capable of spreading the infection. The history of glanders has repeatedly shown that where only the clinical cases were destroyed in a stable or on a ranch, and the remaining contact animals left untouched, the disease would reappear in the form of clinical cases from time to time.

There are several tests that are used for glanders, but the most universally recognized one is the mallein test. Mallein is a biological product prepared in a laboratory from the growth and development of an artificial culture of the germ that causes this disease (*Bacillus mallei*). After a suitable growth of the organism in an incubator, all life in the culture is destroyed by heat, and the dead germs are later filtered out. The filtrate is the substance used in the mallein test. When injected subcutaneously into animals affected with either the external or internal forms of glanders it causes an appreciable rise of body temperature, and quite often considerable edema at seat of injection, malaise, etc. In the hands of a veterinarian who is careful, and with the use of approved mallein, the percentage of errors in diagnosis are exceedingly small.

While an owner in possession of one or more horses or mules affected with glanders should, in protecting his other animals from this disease, be glad to destroy them, it appears to the writer that the destruction of one or more animals affected with this disease is a protection to the entire horse and mule industry of the State. It seems then that some equitable measure should be adopted by this State to reimburse owners for a part of the value of an animal destroyed for glanders. This seems particularly just in cases in which the animals are able to work. This system is in vogue in several of the Eastern States of this country, as well as in the Dominion of Canada. There are two means by which this can be accomplished, viz., either by a direct appropriation for this purpose, or by enacting legislation taxing the horse and mule industry of California for this specific purpose. I am rather inclined to favor the latter. I realize that it would be difficult to obtain a sufficient appropriation from the legislature for this purpose. In the past twelve months the value of horses and mules killed for glanders in California, computing their value at about one hundred dollars per head, was approximately \$70,000.00. If owners were reimbursed a part of the value of destroyed animals, this amount would be raised considerably on account of the fact that many more cases of glanders would be reported to this office directly by the owners, instead, as happens in many instances now, of being kept hidden.

Such a system of reimbursement would in a likewise manner assist greatly in the work of extermination of this disease. While the fund for this purpose would necessarily require to be quite large for the first two to four years, it would only be a comparatively short time when by legislative action the rate of taxation could be lowered on account of the fact that the bulk of the work of glanders extermination would be accomplished during the first few years after this system was adopted.

During the past two years the following outbreaks of glanders were reported. As will be noted the majority of these reports date from the passage of the act of the last legislature requiring the reporting by veterinarians of glanders and other communicable diseases of animals to the State Veterinarian:

	Tested.	Destroyed without test.	Tested and passed.	Held for retest.	Tested and destroyed.	Total number destroyed.
Riverside -----	0	3	0	0	0	3
Tehama -----	17	0	13	0	4	4
Santa Clara -----	8	3	1	0	7	10
Alameda -----	158	27	104	9	44	71
Orange -----	279	18	239	28	12	30
San Diego -----	25	13	17	1	7	20
Madera -----	163	19	107	34	22	41
Mariposa -----	7	2	2	2	3	5
Glenn -----	40	1	6	20	14	15
Kings -----	61	11	27	13	21	32
Kern -----	8	3	0	0	8	11
Santa Barbara -----	13	4	12	1	0	4
Tuolumne -----	1	6	0	0	1	7
Sutter -----	0	5	0	0	0	5
Monterey -----	32	9	24	0	8	17
Marin -----	0	1	0	0	0	1
Placer -----	2	3	0	0	2	5
Trinity -----	0	1	0	0	0	1
San Benito -----	13	2	5	4	4	6
Nevada -----	0	1	0	0	0	1
Stanislaus -----	101	23	83	26	12	35
Amador -----	1	0	0	0	1	1
San Mateo -----	0	1	0	0	0	1
Sacramento -----	30	1	12	5	13	14
Los Angeles -----	158	35	104	15	39	74
Shasta -----	21	0	0	8	13	13
Siskiyou -----	1	0	0	0	1	1
Mendocino -----	5	1	4	0	1	2
San Joaquin -----	168	30	125	26	17	47
Butte -----	161	2	113	23	25	27
San Bernardino -----	6	11	1	0	5	16
San Francisco -----	126	39	67	33	26	65
Santa Cruz -----	8	6	5	1	2	8
Napa -----	37	4	31	3	3	7
San Luis Obispo -----	16	13	6	2	8	21
Fresno -----	158	31	73	59	26	57
Yuba -----	2	2	1	1	0	2
Merced -----	76	20	47	22	7	27
Tulare -----	89	58	48	4	37	95
Totals -----	1,991	409	1,277	*340	393	802

*The 340 animals held for retesting gave doubtful reactions on the occasion of the first test. They were tested a second time, and these tests are included in the 1,991 head of animals tested as indicated in the first column of figures.



Horse in last stages of glanders. Found in a field within 4 miles of Hollister, San Benito County.

SCABIES IN SHEEP.

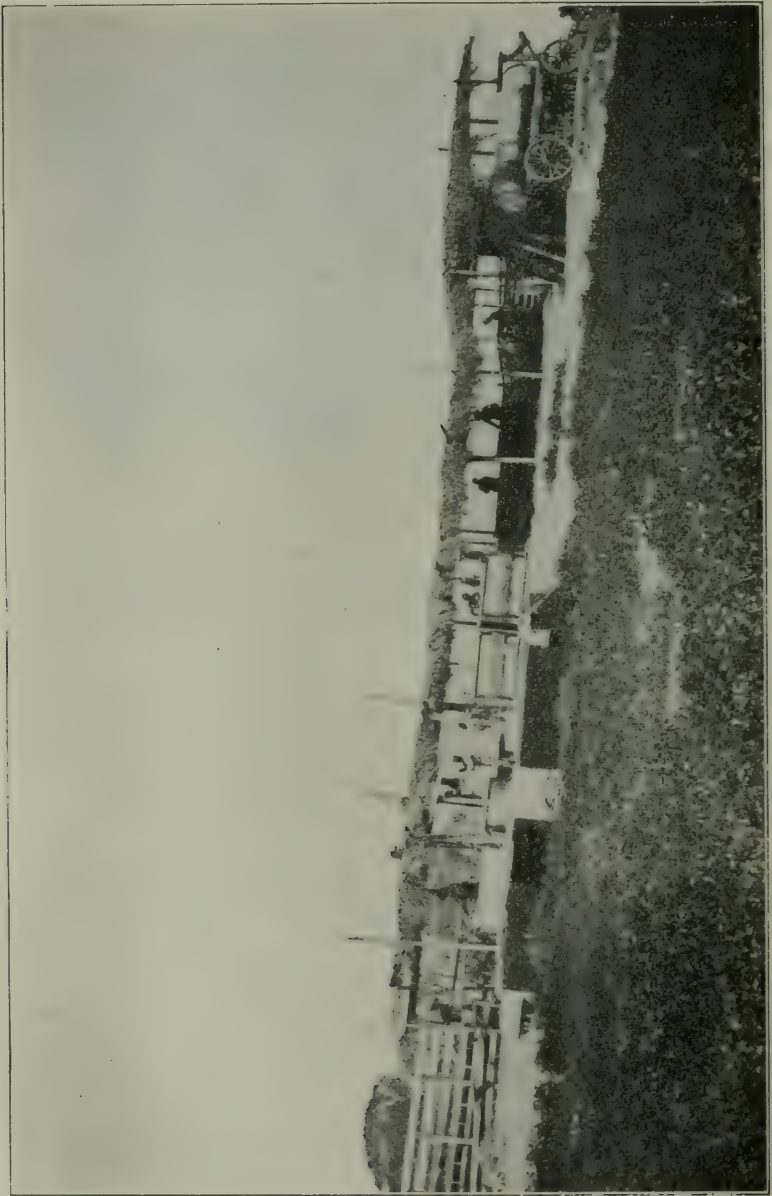
The work of systematic eradication of scabies in sheep, which was inaugurated in March, 1908, has progressed to such an extent that it is only a question of a short time when California sheep will be entirely free from this parasitic disease. When it is taken into consideration that prior to the date on which this work was started practically all of our bands of sheep were more or less infected with scabies, and that the dipping of every band of sheep was officially supervised until at the present writing only isolated foci of infection are found, it can readily be seen what progress has been made in this work. That the result of this work has proved of great benefit to our sheep industry is very apparent to one who was familiar with the conditions that existed several years ago and as they exist to-day.

During the past two years only infected and directly exposed sheep have been dipped. Continual inspections are constantly made by State and Federal inspectors in order to locate as soon as possible any evidences of scabies, and when such sheep are found they are immediately quarantined and dipped under official supervision.

Continued and close inspection of sheep must be made for some time to come. In the driving of sheep to the shearing corrals, and in the annual drives to and from the summer ranges, there is always the danger of spread of scabies if one infected band is overlooked.

The act of February 23, 1909, giving the State Veterinarian authority to seize and dip a band of sheep infected with scabies in case the

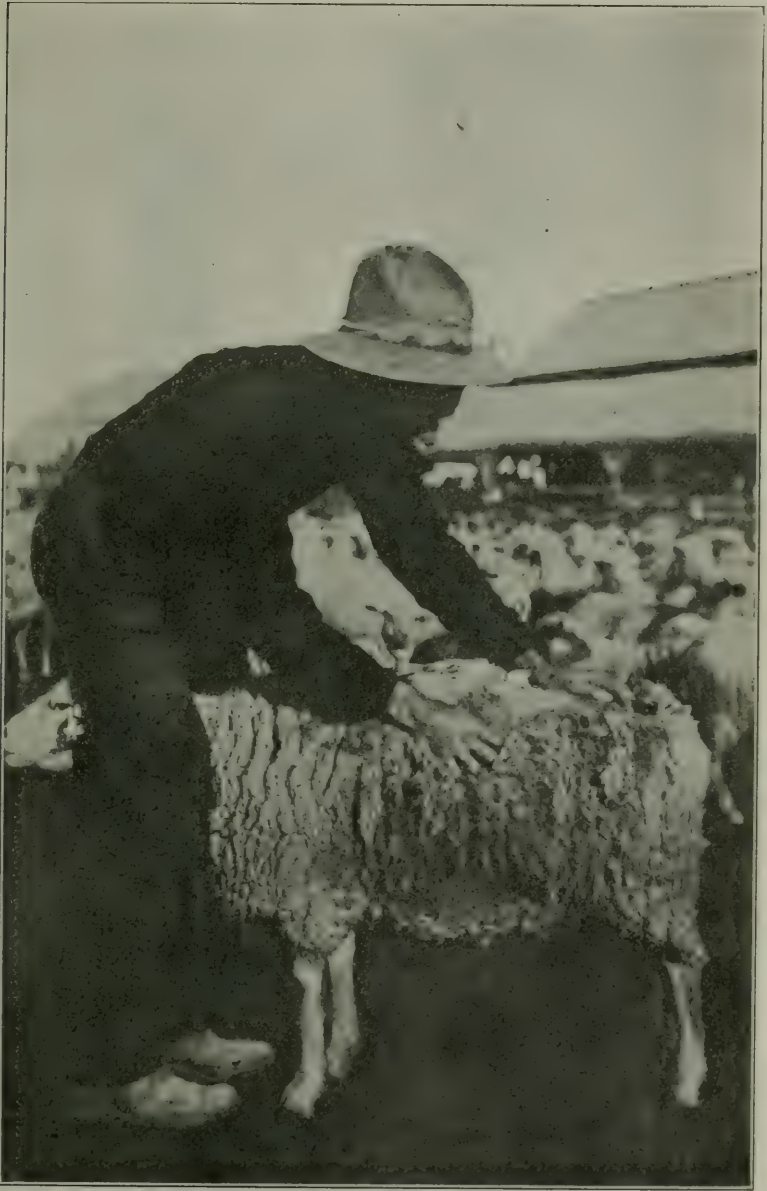
owner refused or neglected to do so, has assisted this department greatly in the conduct of this work. Prior to the enactment of this



A large sheep dipping plant located in the San Joaquin Valley.

law our authority was practically limited to quarantine measures. While we have found it necessary to enforce the provisions of this act only in a few instances, in a number of other instances notices to

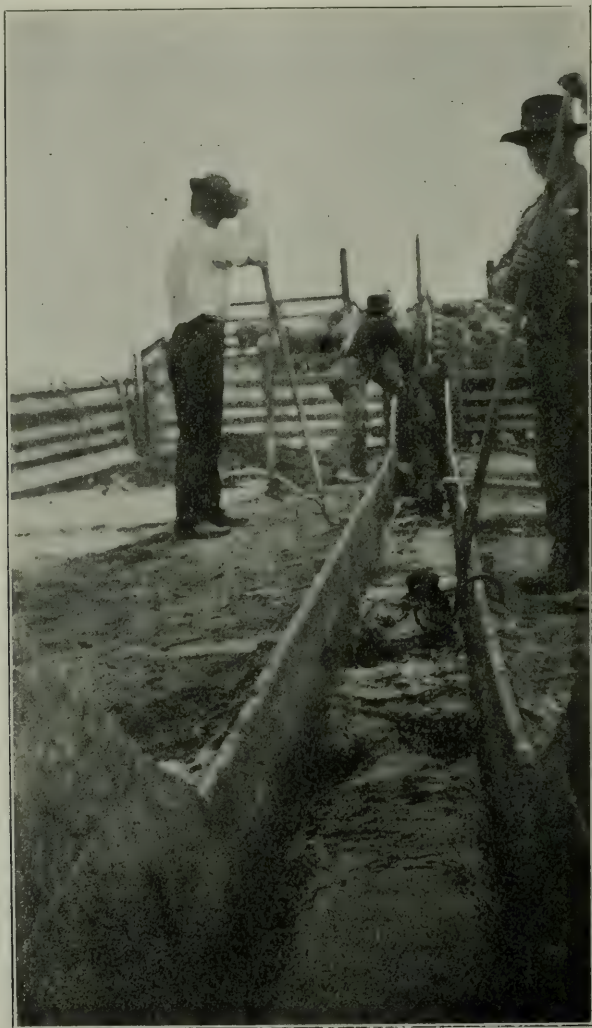
dip sheep have been served by our inspectors under the provisions of the act, with the result that the owners finally consented to comply



Sheep infected with scab, showing loss of wool.

with our instructions before the ten days' notice to dip had expired. In 1909 one band of sheep was seized and dipped in Alameda County, the owner paying the costs of such dipping without the necessity of

the State resorting to the courts. In 1910 another band of sheep was seized and dipped by the State in San Benito County, the expenses of which were paid by the owner of the sheep.



Dipping sheep for the eradication of scabies on a California sheep ranch.

During 1910 two arrests were made in Fresno County for shipping scabby sheep. In one case the guilty party was fined. The other case was dismissed through lack of evidence.

RABIES.

In the fall of 1909 reports were received at this office of dogs with rabies (hydrophobia) in the city of Los Angeles. Later these reports came with alarming frequency, not only from Los Angeles, but also from Pasadena, Stockton, and Riverside. The city veterinarian of Pasadena was requested to remove the heads from rabid dogs, pack same in ice, and send them to the State Hygienic Laboratory for the purpose of having the diagnosis verified by microscopical examination. These examinations were conducted by a veterinarian connected with the Pathological Laboratory of the United States Bureau of Animal Industry who had had considerable laboratory experience with the technique necessary for the laboratory diagnosis of rabies. He was detailed to conduct these examinations by the Chief of the United States Bureau of Animal Industry at the request of the State Veterinarian.

The following is the report made by Dr. Hart of the heads submitted for examination:

"I beg to submit to you the following report of my investigations during the time I was detailed at the California Hygienic Laboratory at Berkeley, Cal., November 18th to December 6, 1909.

The object of the work was to prove or disprove the supposed existence of rabies in the State of California. I arrived at the laboratory on November 19, 1909.

The brains of two dogs (Laboratory Nos. 5 and 6) which had been affected with the suspected disease had arrived that day. They were from Pasadena, Cal., and were forwarded in glycerin by Dr. W. A. Boucher, city veterinarian of Pasadena.

The stain in Mann's method of demonstrating Negri bodies was not in stock in the hygienic laboratory and could not be procured in San Francisco. I was, therefore, obliged to use the alcoholic eosin and methylene blue in Frothingham's method. In sections of the hippocampus major of dogs Nos. 5 and 6 stained in this manner, suspicious bodies in the nerve cells were found at once, but no preparation sufficiently clear to make a diagnosis could be secured. It was, therefore, requested that the entire head of the suspected cases be forwarded in order that the plexiform ganglia could be removed and the changes of Van Gehuchten and Nelis demonstrated. In response to this request Dr. Boucher forwarded two dogs' heads (Laboratory Nos. 7 and 8), which were received on November 21st. In No. 7 poorly stained Negri bodies and well-marked ganglionic changes were found.

On November 24th two more dogs' heads (Laboratory Nos. 9 and 10) were received from Pasadena, and on the same day the stains in Mann's method for Negri bodies, which had been requested by telegram from the Pathological Division of the Bureau, arrived. With this stain very clear preparations of Negri bodies were found in sections of the hippocampus major of dogs Nos. 5, 6, 7, 9 and 10. The histologic changes in the ganglia were also very typical in dogs 9 and 10, and the diagnosis of rabies was reported to the State Veterinarian, to W. A. Boucher and to Dr. W. M. MacKellar on November 26th.

The brains of dogs (Laboratory Nos. 1, 2, 3 and 4) had been forwarded before my arrival, and Dr. A. R. Ward in charge of the laboratory had inoculated two rabbits subdurally with an emulsion of each of these brains. On November 27th and 28th the rabbits from dogs Nos. 2 and 3 showed symptoms of rabies, and on November 29th both rabbits from dog No. 2 and one from dog No. 3 were dead of the disease. The hippocampi of these experimental animals were removed and Negri bodies demonstrated in both cases.

On November 27th Negri bodies were found in the brain of a dog (Laboratory No. 11) from Los Angeles forwarded by Dr. J. R. Shaw in formaldehyde. On November 29th a dog (Laboratory No. 12) was received from Stockton, Cal., for examination. The microscopic examination of this case was inconclusive, and experimental inoculations were made. Rabbit inoculations had been made in all cases,

however, but after those inoculated by Dr. A. R. Ward developed the disease, it was not considered necessary for me to remain longer, and I left Berkeley on December 4th.

The existence of rabies was demonstrated in eight dogs. This is the first positive finding of the disease in this State, although it has undoubtedly existed here for several months at least, and California will now have to be added to the list of States in which rabies prevails.

Very respectfully,

[Signed.] GEORGE H. HART,
Assistant in Pathology and Bacteriology."

Dr. Hart's examination not only confirmed the physical diagnosis made in Los Angeles, but also a microscopical examination made in the laboratory of the city board of health of that city. Since that time numerous examinations have been made in Los Angeles and in the State Hygienic Laboratory, and in the majority of instances these examinations have been positive.

Not only were reports received at this office of the existence of this disease among dogs, but reports of rabies in human beings, horses and cats were also received. From southern California two deaths in human beings from rabies were reported, one a child and the other an adult.

Reports received from veterinarians in Stockton of rabies in dogs amounted to over 100 cases. Several hundred cases in dogs were reported by veterinarians in the environs of Los Angeles.

As rabies is, under natural conditions, only transmitted from one animal to another, including man, through infected saliva by biting, this department endeavored, in coöperation with the municipal authorities in Stockton, Los Angeles, and Pasadena, to have ordinances enacted requiring the muzzling of all dogs when the animals were not confined on the premises of their owners. Such ordinances were enacted, but were, in the opinion of the writer, repealed too quickly. The nature of the disease, with its variable period of incubation, requires the enforcement of a muzzling ordinance a matter of months before one can be assured with any degree of certainty that the disease is under control. However, the superstition and ignorance in connection with this disease that has been handed down for ages, has resulted in considerable opposition to the muzzling of dogs.

In view of the fact that rabies has now become endemic in California, it might be well to cite in this report, a few facts regarding this disease, especially as we realize there still remain people who are skeptical concerning, and even deny the existence of rabies as a distinct disease.

RABIES (HYDROPHOBIA).

This disease existed and was recognized before the beginning of the Christian era. It was described by Aristotle and other writers. Experiments in the virulence of saliva from rabid animals were first successfully carried out by Zinke in 1804.

The modern study of this disease may be said to date when Pasteur devised his method of subdural inoculations of material from the central nervous system. This method of diagnosis is very certain and is still practiced to-day as a reliable means of settling the diagnosis beyond a doubt.

Rabies was first observed in America in the latter half of the eighteenth century. It appears to have gradually spread from the north Atlantic States, where it was first observed, to the south and west until at the present writing only a few of the Western States appear to be free from this disease.

ANIMALS SUSCEPTIBLE.

Practically all warm blooded animals are susceptible to rabies infection. The disease is more commonly met with, however, in dogs and other members of the canine family. This is of course due to the natural instinct of the dog to bite. While in man only a relatively small number contract rabies after being bitten it would appear that he possesses a certain amount of resistance to the disease. The relatively few cases in man after bites by rabid dogs, can, however, possibly be explained in several ways. In the majority of instances these bites are through the clothes, which to a certain extent prevents the introduction of the infected saliva into the wound. Immediate cauterization of the wound is also universally practiced after dog bites. On the contrary, in dogs very little protection is offered from bites, except by the hair, and little attention is paid to these bites afterwards.

CAUSE OF RABIES.

In practically all cases rabies is caused by the bite of a rabid animal. The rabid dog is the principal source by which rabies is spread. While from the nature of this disease and experimentation, we are assured it is infectious, still the exact nature of the organism is not definitely understood. The incubation period, or the time that elapses between the bite and when the symptoms of rabies appear, is remarkable for its variability. It is seldom that the disease appears in nature before the tenth day, while the longest period may run ten to twelve months. The majority of cases occur, however, before the third month.

SYMPTOMS IN DOGS.

The disease manifests itself in two distinct types in dogs, viz., the furious and the dumb type. A very characteristic early symptom is a marked and apparently causeless change in the natural disposition of the dog. Pet animals, which were fondled, appear suddenly to become morose and irritable, while those which formerly were naturally irritable may show unusual affection, and desire to be fondled and petted.

The initial symptoms of the rabid dog are restlessness. He is constantly moving. He barks and growls on slight provocation and is easily startled. In the furious type, if the dog is not restrained, he may suddenly leave home and wander off a great distance to return home again in a day or two greatly changed. It is during this wandering period that the rabid dog usually bites people and other animals. While he does not appear to go out of his way to attack men or animals he resents any interference with his movements, and does not hesitate to use his teeth in such resentment. There appears to be hypersensibility of the skin as is evidenced by the animal biting itself. He appears to have no appetite for food, but will frequently bite and swallow indigestible material, such as wood and stones. The act of swallowing becomes difficult, and later in the disease, is impossible.

In the latter stages of the disease convulsions appear and it sometimes happens that death occurs in one. Frequently, however, death occurs from paralysis. The rabid dog finds a secluded place. The lower jaw drops from paralysis of its muscles. The symptoms of paralysis extend until death occurs from paralysis of the centers of respiration.

Another characteristic symptom often noted by those in attendance upon rabid dogs is a change in the character of the voice. Instead of the natural succession of sharp barks there is a hoarse howl followed by an unequal series of barks.

Dumb rabies is distinguished from the furious type by the brevity or complete absence of the stage of excitement. The early symptoms exhibited are those of paralysis of the muscles of deglutition (swallowing, and the muscles of the lower jaw).

The common fallacy that rabid dogs are afraid of water, is due to this throat paralysis. While the dog is thirsty it is unable to swallow.

DIAGNOSIS.

This is determined by the symptoms, cause and termination of the disease, microscopical examinations, and inoculation of experimental animals.

While the symptoms and cause of rabies appear to be sufficient for one experienced with this disease to render a diagnosis, it is well to confirm this diagnosis by laboratory methods. This microscopical examination consists in the search of the brain for the microscopical bodies (Negri bodies) discovered by Negri in 1903, and which, when found, are to-day considered as a positive diagnosis. These bodies can only be detected in the brains of animals which are affected with rabies.

Other microscopic lesions have some value in confirming a diagnosis among which might be mentioned the ganglionic changes of Van Gehuchten and Nelis, as well as other minute lesions found by other investigators.

Rabbits inoculated subdurally usually develop the disease by the third week.

SUPPRESSION OF RABIES.

The measures by which rabies can be exterminated are well known. Unfortunately many fallacies regarding this disease have been handed down to us, and its existence is often denied in face of indisputable facts. In some of the European countries little difficulty was experienced in eradicating this disease where suppressive measures were properly administered. The disease does not exist in Australia, and is not likely to be introduced into that country, on account of the quarantine measures in force to prevent its introduction.

As the dog is the natural disseminator of rabies, measures to prevent the dissemination of the disease by these animals, should be adopted and rigidly enforced. Among such measures might be mentioned the destruction of all ownerless dogs, the licensing of all dogs and the muzzling of all dogs when not confined within the premises of their owners.

On account of what we know regarding the variability of the incubation period of rabies, muzzling should be abandoned only after a sufficient time has elapsed, as to warrant a feeling of security that the disease has been eradicated. In the opinion of most observers muzzling should be in force for not less than six months.

ANTHRAX.

This disease has continued to be the cause of severe losses to our live stock during the past two years, as it has been in former years. As long as live stock are run on infected pastures, just as sure will fatalities occur from this disease. Once infected a pasture remains infected indefinitely, and as soon as the feed in such a pasture becomes short, animals start to die. This is probably due to the fact that infection exists close to the grass roots. This fact is so well recognized on some of our larger beef ranges, on which in former years anthrax was the cause of severe losses, that the infected range is not allowed to be fed off too close. The cattle are taken off early and only put back after a new growth of feed appears.

On many of the smaller ranches this procedure can not be carried out on account of lack of feed. On these places, in most instances, preventive vaccination is attempted with variable results. In some instances cattle continue to die after vaccination, while in others vaccination when properly performed, appears to check the progress of the disease.

Anthrax is a very serious disease, especially when it occurs in a dairy herd. It is directly communicable to man, who may possibly become infected through the milk from an infected cow. The disease appears

very suddenly as a rule, one or more cows being found dead which a short time previous were apparently enjoying good health. While any disease in a milch cow that causes a high fever will eventually suppress milk secretion, there is a period, however, during the early stages of the disease during which milk is secreted. Grave danger exists if such milk is used for human consumption. On account of the virulence of this disease and its suddenness in attack, measures are adopted by this department to prohibit the use for any purpose of milk from any cows in a dairy where anthrax exists. Such milk is ordered destroyed immediately after milking, and until such a time as it would appear that the disease has abated.

During August and September of 1909 serious outbreaks of anthrax occurred among the cows in four dairies in Solano County. Two hundred and three cows died from this disease in these four dairies in the course of about two weeks.

The following reports of outbreaks of anthrax were investigated by this department during the past two years: Kings County, on three ranches; Merced County, on four ranches; San Benito County, on three ranches; Stanislaus County, on one ranch; Kern County, on one ranch; Contra Costa County, on six ranches; Glenn County, on one ranch; Colusa County, on one ranch; Solano County, on seven ranches; Yolo County, on two ranches; Inyo County, on eleven ranches; Monterey County, on three ranches; Tehama County, on five ranches; Sacramento County, on two ranches; San Joaquin County, on one ranch; Sonoma County, on one ranch.

The above does not come near representing the outbreaks of anthrax during the past two years. In many instances deaths among live stock due to anthrax occur on the ranges along our large river courses and no notification is received at this office regarding same.

HOG CHOLERA.

Numerous reports of deaths in hogs from cholera were received and investigated by this office. In the following counties the losses from this disease were quite extensive: San Luis Obispo, Fresno, Los Angeles, Mendocino, Kings, Tehama, Santa Cruz, San Joaquin, Tulare, Santa Clara, Monterey, Merced, Colusa, and Shasta. Hog cholera is, however, pretty well distributed throughout the hog raising sections of this State, and acts not only as a menace to hog raising, but also prevents the natural development of this important industry. California is particularly adapted for the raising of hogs. The large areas in this State which have in the past few years been turned into dairies through the development of irrigation, offer an excellent field for the production of pork. The by-products from these dairies, as well as the class of feed that can be grown under the system of irriga-

tion, insures the production of pork at a relatively low cost. To what extent hog raising can be reached in California, is only limited by the protection that can be afforded hog raisers against the ravages from hog cholera.

At the present time we raise a very small part of the pork that is consumed in this State. Nearly all of our pork is brought into California in refrigerator cars in sweet pickle from the packing centers of the Middle West. For fresh pork many car loads of hogs are purchased by California packers in other states and hauled out here for slaughter. Why this condition should exist in an agricultural state like California, where every facility exists for hog raising, sufficient to supply at least our own consumption of pork, is undoubtedly due, in the opinion of the writer, to the fact that a hog breeder is afraid to take chances with cholera decimating his herd. This opinion is shared by many others in this State. The deaths from this disease are so rapid that under present conditions in California it spreads rapidly from one animal to another until in a very short time practically the entire herd is dead.

At the present time in California our only means of fighting against the ravages of cholera is by the ordinary sanitary procedure. We supervise the cleaning and disinfection of premises where the disease exists, cremation of carcasses, and advise the dipping of all contact hogs in a disinfectant.

In the fourth biennial report of this department I called attention to a new method of preventive vaccination against cholera in hogs, as adopted and discovered by the Bureau of Animal Industry, United States Department of Agriculture. The Department of Agriculture has patented this process in such a manner as to insure all the people of this country the right to its free use. The department, however, does not distribute this serum to the states or to hog raisers. The various states are expected, if they wish to protect their hog industry by this method, to prepare and distribute this serum.

METHOD OF SECURING IMMUNE SERUM.*

Without attempting to go into the method of producing this serum in detail, it will be sufficient to say that the protective serum is produced by a process of "hyperimmunization" carried out as follows: An immune hog is injected with large amounts of blood from hogs sick of hog cholera. These injections will not produce more than a transitory effect upon the health of the immune, although they would prove certainly fatal to a susceptible hog. This treatment of immune hogs with large amounts of disease-producing blood is known as "hyperim-

*Excerpt from a paper read by Dr. A. D. Melvin, Chief of the U. S. Bureau of Animal Industry, at the forty-fifth annual meeting of the American Veterinary Medical Association.

munization," and gives to the blood of the immune the power to protect susceptible hogs from hog cholera. After a week or so, when the immune has recovered from the effects of this treatment, blood is drawn from the immune by cutting off the tail. The blood-drawing is repeated three or four times, at intervals of a week between the drawings, after which the immune is usually bled to death from the carotid. After each drawing from the immune the blood obtained is defibrinated and mixed with a suitable antiseptic. If preserved in sterile bottles this defibrinated blood, or serum, as it is called, will retain its potency for years.

The protective serum having been obtained from an immune hog in the manner indicated, the potency of this serum is determined by injecting susceptible pigs with varying amounts of the serum and at the same time exposing them to hog cholera along with untreated control animals. In practice it will of course be found best to first collect large quantities of serum and to mix these before testing. A standard serum will thus be secured at a minimum cost.

METHOD OF PROTECTING SUSCEPTIBLE PIGS.

A standard serum of known potency having been secured, either of two methods may be used for protecting susceptible pigs. These are known as (a) the serum-simultaneous method, and (b) the serum-alone method.

The first of these, which is to be recommended for use especially in herds which have not been exposed to hog cholera, consists in injecting subcutaneously on one side of the body of the pig to be vaccinated a suitable quantity of serum and simultaneously on the other side of the body a small quantity of virulent blood taken from a hog sick of hog cholera. Experiments have shown that by this method pigs are given a firm immunity, lasting at least six months and probably much longer.

The "serum-alone method," which consists simply in the injection of the protective serum without the simultaneous use of disease-producing blood, appears to confer only a temporary immunity upon the treated pigs, unless they be exposed to hog cholera a short time after receiving the serum, in which case they also acquire a lasting immunity. For these reasons the "serum-alone method" is admirably adapted to the treatment of hogs in a herd where hog cholera has already broken out, but which have not themselves shown visible symptoms of disease.

The experiments which are being carried out to determine the curative properties of the serum are not yet complete, but from the results thus far obtained we know that serum in the doses used for immunization can not be depended upon to cure hogs which already show visible

symptoms of hog cholera. Further work along this line is needed. It should be stated that neither the serum-simultaneous nor the serum-alone method, when properly applied, appears to injure the hog in any way.

COST OF THE SERUM.

In order to determine the cost of producing serum for practical use every item of cost would, of course, have to be taken into account and allowance made for all sources of revenue. Owing to the conditions under which we have been working, that is, manufacturing serum for experimental use only, and utilizing the same force for the production of the serum and for carrying on varied experiments, we are unable to give an exact cost price of the serum thus far produced. Sufficient work has been done, however, for an estimate to be made. With the dose of serum at 20 cubic centimeters, and with the production carried out with strict economy, it seems likely that the cost per dose can be brought to 25 cents. This estimate is based upon the supposition that each hyperimmunized immune will furnish 150 to 200 doses of serum, and that the carcass of the immune after final bleeding, will be utilized for food. There seems to be no objection to the use of such a carcass for food purposes, provided the post-mortem examination discloses no reason for rejecting it.

I have recently been informed by Dr. C. E. Marshall, of the Michigan Agricultural College, who has begun the production of this serum for distribution to farmers of that state, that it is their purpose to charge at present 2 cents per cubic centimeter for the serum, though they hope to be able to reduce the price materially before another season. It will undoubtedly prove to be true that the serum cost will vary with the conditions of manufacture, and the proportionate cost should decrease as the amount of serum produced increases.

In any case it seems certain that the serum can be produced cheaply enough for practical purposes.

RESULTS OF PRACTICAL TESTS OF THE SERUM.

The statements which have been made above concerning the protective power of serum from hyperimmunized immunes are based upon tests upon several thousand hogs, and these tests were not carried out in small experiment pens only but in great part upon farms under practical conditions. During the fall of 1907 approximately 2,000 hogs were treated on fifty different farms, a considerable proportion of untreated hogs being left in all cases as a control on the action of the serum. Both methods of vaccination were used and the herd conditions varied widely. The herds can be roughly classified as (a) those in an infected district but themselves free from disease; (b) those which were known to have been exposed by contact with sick hogs but

which had not developed disease at the time of treatment, and (c) herds in which hog cholera was present and hogs sick and dying at the time of treatment.

In no cases were any of the ordinary methods of combating hog cholera by disinfection and separation of the sick from the apparently healthy practiced. Where disease was present at the time of treatment, the treated were allowed to run with the sick animals along with a number of untreated animals which served as controls, and the success following vaccination can therefore be attributed to the action of the serum. In the herds where hog cholera appeared subsequent to treatment, all of the vaccinated hogs remained well, while more than 65 per cent of the checks died. In the herds which had been exposed but were apparently well at the time of treatment 4 per cent of the treated animals died, while approximately 90 per cent of the checks succumbed. In the herds where disease existed at the time of treatment, and where we did not anticipate very great success, 13 per cent of the treated animals were lost, whereas 75 per cent of the checks died.

These successful field trials, confirming as they did numerous tests carried out under experimental conditions, have convinced us of the efficiency of the method of dealing with hog cholera; and although improvements will undoubtedly be made in many of the details of producing the serum, the method is believed to be now in such condition as to make the practical use of it entirely feasible.

In order that the states most concerned in this question might be brought into closer touch with the work, and also for the purpose of discussing plans for effective coördination of the State and Federal work in dealing with hog cholera, twenty-five of the chief hog raising states were requested to send representatives to Ames, Iowa, where the Bureau of Animal Industry maintains a farm devoted to experiments with hog cholera. In response to this invitation representatives from twenty different states visited Ames and were shown the details of the serum production.

A general discussion at these conferences developed the practically unanimous opinion on the part of State and Federal representatives that the serum should be prepared by each of the states for distribution to the hog raisers, and all State representatives expressed their intention to undertake the work as soon as funds could be secured. At the present time a number of states have actually begun work. If the serum is prepared in sufficient quantities, there seems to be no doubt that a great saving can be effected simply by treating animals in exposed herds, or in herds in which the disease has just appeared; if the greatest good is to be accomplished, however, we should not be content simply to reduce the losses from hog cholera, but should under-

take systematic efforts to eradicate the disease. We are all familiar with the course which hog cholera usually takes when it appears in a neighborhood. A herd develops the disease which may not be recognized as hog cholera for several weeks after it has made its appearance. Following this the neighbors' hogs will become infected, and from there rapid progress is made, the number of new foci increasing more rapidly as the infected area widens, until finally the losses in a single county may be enormous.

It is evident that in order to control the disease there must be some means of confining it to the original center of infection. This has been attempted by the British Government through the quarantine of farms where hog cholera exists, and the slaughter of all infected animals. That such procedure alone will not yield the desired results is shown by the official reports of the prevalence of hog cholera in England.

In this country such methods would not be suitable, for, aside from the enormous expense involved, it would in my opinion be entirely impracticable to thoroughly disinfect extensive farm premises and to carry out a quarantine which would be effective against such carriers of disease as dogs, crows, buzzards, and other animals.

While the production of this serum appears to be a very simple process, suitable buildings and equipment are necessary, including a laboratory. The experience of officials in several of the states in the Middle West, engaged in the production of anti-hog cholera serum, has shown that the cost of production is about 30 cents for an average dose. The plan adopted by these states is to provide proper equipment and help and supply the serum to hog raisers at the cost of production.

By the establishment of such a plan in this State, our hog industry would not only receive the protection it requires, but it would also be built up to that stage where in time it would become a very profitable business.

BLACK LEG (SYMPTOMATIC ANTHRAX).

This disease, which affects young cattle, is prevalent to a greater or less extent throughout our cattle raising sections. It is easily controlled, however, by preventive vaccination, which is universally practiced by cattle raisers. A number of outbreaks were reported to this department by cattle raisers who were ignorant of the nature of the disease. These reports were investigated and the nature of the diseased established, and preventive vaccination was resorted to with the result that the disease was checked. Cattle raisers are advised by this department, when once the disease appears in the herd, to annually vaccinate their young stock.

FORAGE POISONING IN HORSES AND MULES.

The losses from this affection in several sections of the State were very serious. This affection was investigated and relief afforded in Siskiyou, Santa Clara, San Joaquin, and Fresno counties. It is due to the eating of spoiled or mouldy fodder, and is very fatal to horses and mules. The affection has in the past been frequently diagnosed as epizootic cerebrospinal meningitis, by reason of the fact that there is an involvement of the central nervous system as is evidenced by the train of symptoms. The affection is easily prevented by a complete change of the animals to bright, clean fodder.

TUBERCULOSIS.

In the last biennial report of this department I drew attention to the fact that tuberculosis among hogs in California was increasing to an alarming extent. The reports received at this office during the past two years from the Federal meat inspectors of tuberculosis found in hogs at official abattoirs were very numerous. That hogs contract this disease from tubercular cattle there is no question of doubt. Experimental and clinical evidence conclusively proves this fact.

Tuberculosis is a highly infectious disease and prevails to an alarming extent among the dairy cows of California. This fact is generally admitted by our most progressive dairymen and cattle breeders. In fact, the most universal prevalence of this disease among dairy stock is the principal argument presented against measures that may be adopted for its suppression. However, some plan should be adopted by this State looking towards the ultimate suppression of this disease. Tuberculosis is constantly spreading among cattle and hogs, and it is only a question of time when the public will demand that measures be taken to cope with this disease.

This subject has a public health as well as an economical interest. The fact is well established that man can contract tuberculosis from the ingestion of milk or meat from infected cows; and tuberculosis, studied from an economical view-point is the cause of severe losses to our cattle and hog industries. It not only shortens the lives of cows, but unquestionably reduces the efficiency of the herd as a machine for producing milk. The disease is also very prevalent among the cattle on some of our beef ranges. It is the cause of death on the ranges, and severe losses are experienced through the condemnation of carcasses in official abattoirs by Federal inspectors.

The control of tuberculosis in animals is a state problem. It has been attempted by municipalities in California in the enactment of ordinances for the improvement of milk supply without much success in any instance. The State should endeavor by adequate legislation to ascertain in what herds this disease exists, and on gaining this infor-

mation the tuberculous animals should be segregated from the balance of the herd. At the present time the only practical method of obtaining this information is by the application of the tuberculin test to all dairy cattle in the State of California.

In order to guard against its illicit use, legislation should be enacted regulating the sale of tuberculin, and persons who intend to use the same should be required to obtain permits from the authorities charged with the control of animal diseases. Legislation of this nature would be imperative if any attempt should be made by the State to segregate the infected from the healthy animals. The practice of plugging, or in other words, the injection of tuberculin into a cow a short time before the inspector comes around to make the test is pretty well understood. Under such circumstances a tuberculosis cow will not react to tuberculin for some time, having by the first injection developed a toxic immunity to same.

The states of Kansas, Wisconsin, and New York have enacted legislation regulating the use of tuberculin and requiring persons who intend to use the same to obtain permits from the authorities charged with the control of animal diseases.

The writer fully realizes the magnitude of the work of suppressing tuberculosis in our cattle herds. The work should be started, however, and the proper authorities should be supplied with adequate facilities and wise legislation in order to carry on this most important work.

LABORATORY REQUIREMENTS.

This department has got along as best it could since its establishment by the legislature of 1899 without any laboratory assistance. The modern study of communicable diseases of animals, having for its principal object the control and eradication of these diseases is accomplished both in the field and in the laboratory. One is a necessary adjunct to the other. The need of such a laboratory in connection with this department is felt. No department charged with the control and eradication of infectious diseases can be expected to do effective work without such a laboratory. In addition to the investigation of diseases, this laboratory could prepare and distribute hog cholera serum as indicated by my remarks on that subject.

I would, therefore, recommend the establishment of such a laboratory, and the creation of the position of veterinary bacteriologist and pathologist in connection with this department.

SOUTHERN CATTLE FEVER QUARANTINE REGULATIONS ISSUED DURING THE SIXTIETH AND SIXTY-FIRST FISCAL YEARS.

STATE OF CALIFORNIA,
OFFICE OF STATE VETERINARIAN.

PROCLAMATION.

EXECUTIVE DEPARTMENT, STATE OF CALIFORNIA,
SACRAMENTO, CAL., June 14, 1909.

WHEREAS, The State Veterinarian of the State of California has ascertained that cattle located in certain counties in the State of California hereinafter named are liable to communicate an infectious disease known as Texas, splenic or Southern fever, to cattle located in other counties in the State of California should said cattle located in the counties hereinafter named be shipped, moved, transported, driven or grazed over the lands situated in other counties in the State of California; and,

WHEREAS, Under and by virtue of an act of the legislature of the State of California, entitled "An act to protect domestic live stock from contagious and infectious diseases, to provide for the appointment and duties of officials to carry into effect the provisions of this act, and to provide an appropriation therefor," which became a law March 18, 1899, and as amended March 20, 1905, and March 23, 1907, and as further amended March 19, 1909, the State Veterinarian of the State of California, in order to prevent the spreading or communication of said disease of Texas, splenic or Southern fever in cattle, has, on this fourteenth day of June, 1909, quarantined the following counties and parts of counties in the State of California: San Luis Obispo, Santa Barbara, Ventura, Los Angeles, Orange, San Diego, Tulare, and Kings; and that part of Fresno county situated west of the east side line of the Southern Pacific Railroad; and those parts of San Bernardino and Riverside counties situated west of the one hundred and sixteenth meridian, west longitude.

WHEREAS, The State Veterinarian has, on this fourteenth day of June, 1909, made and established the following rules and regulations as to the movements of cattle from said counties and parts of counties into other counties or sections in the State of California, or from any one of the counties or parts of counties herein mentioned to any of the other counties or parts of counties herein mentioned.

From and after this fourteenth day of June, 1909, every person, company, corporation, their agents and servants, are hereby prohibited from driving, leading, moving, carrying or transporting, or from causing or permitting to be driven, led, moved, transported, carried or drifted into any other county or section in the State of California, or from any one of the counties or parts of counties herein mentioned, to any of the other counties or parts of counties herein mentioned, any cattle originating or being in the following named counties and parts of counties: San Luis Obispo, Santa Barbara, Ventura, Los Angeles, Orange, San Diego, Tulare, and Kings; and that part of Fresno county situated west of the east side line of the Southern Pacific Railroad; and those parts of San Bernardino and Riverside counties situated west of the one hundred and sixteenth meridian, west longitude, unless such cattle have first been inspected by the State Veterinarian or his duly authorized deputy and are accompanied by a certificate issued by such officer stating that said cattle are free from contagious and infectious diseases and cattle ticks (*Margaropus annulatus*); provided, however, that when said cattle are shipped or transported on railroad trains for immediate slaughter to the following named cities in the State of California: San Francisco, Oakland, Los Angeles, and San Diego, inspection and certification of such cattle will not be required, but when so shipped or transported by railroad the following regulations must be observed.

On unloading said cattle at their destination, or for food and rest en route to same, separate pens must be set apart to receive them and no other cattle shall be admitted to said pens. The cars that have carried said cattle shall be cleaned and disinfected before they are again used to transport, store or shelter animals or merchandise.

All cars carrying said cattle shall bear placards stating that said cars contain "Southern cattle," and each of the way bills of said shipments shall have a note on its face with a similar statement.

The cars used to transport such cattle shall be cleaned and disinfected in the following manner:

(a) Remove all litter and manure. This litter and manure must be stored where no cattle can come into contact with it for a period of at least eight months.

(b) Wash the cars with water until clean.

(c) Saturate the entire inner surface of the cars, including the inner surface of the car doors, with a mixture made of one and one-half pounds of lime and one-quarter pound of pure carbolic acid to each gallon of water; or with a solution made by dissolving four ounces of chloride of lime to each gallon of water.

It is further ordered that a violation of any or either of the foregoing rules and regulations shall be an offense, and punishable as provided by the laws of the State of California.

Now, therefore, I, J. N. Gillett, as Governor of the State of California, by virtue of the authority vested in me by law, do hereby proclaim the counties and parts of counties named in this proclamation quarantined within the State of California, and further hereby proclaim the foregoing rules and regulations prescribed by the State Veterinarian for the maintenance and enforcement of such quarantine to be legal and binding rules and regulations within the State of California, and I do further proclaim that said rules and regulations shall be maintained and enforced within the State of California, and that a violation thereof shall subject all persons so violating any of said rules or regulations to the penalties provided for in section eight of that said act of the legislature entitled "An act to protect domestic live stock from contagious and infectious diseases, to provide for the appointment and duties of officials to carry into effect the provisions of this act, and to provide an appropriation therefor."

In witness whereof, I have hereunto set my hand and caused the Great Seal of this State to be hereunto affixed, this fourteenth day of June, A. D. 1909.

[Signed] J. N. GILLETT,

Governor of the State of California.

Attest:

[Signed] C. F. CURRY,
Secretary of State.

[SEAL]

STATE OF CALIFORNIA.
OFFICE OF STATE VETERINARIAN.

PROCLAMATION.

EXECUTIVE DEPARTMENT, STATE OF CALIFORNIA,
SACRAMENTO, CAL., March 1, 1910.

WHEREAS, The fact has been determined by the Secretary of the United States Department of Agriculture that an infectious disease known as splenic, Southern or Texas fever, exists among cattle in the following named states, to wit: Oklahoma, Texas, Missouri, Arkansas, Louisiana, Mississippi, Tennessee, Alabama, Virginia, North Carolina, South Carolina, Georgia, and Florida; and,

WHEREAS, Under and by virtue of an act of the legislature of the State of California, entitled "An act to protect domestic live stock from contagious and infectious diseases, to provide for the appointment and duties of officials to carry into effect the provisions of this act, and to provide an appropriation therefor," which became a law March 18, 1899, and as amended March 20, 1905, and March 23, 1907, and as further amended and approved March 19, 1909, the State Veterinarian of the State of California, in order to prevent the spreading or communication of said disease of splenic, Southern or Texas fever to cattle within the State of California, should cattle from the aforesaid states be imported into the State of California, has, on this first day of March, 1910, made and established the following rules and regulations as to the importations of cattle from the aforesaid states:

From and after this first day of March, 1910, every person, company, corporation, their agents and servants, are hereby prohibited from bringing into the State of California any cattle originating or being in that area of territory in the United States south of the Federal quarantine line which has been or may be established

by the United States Department of Agriculture, or any cattle infested with the *Margaropus annulatus* tick, except in accordance with the regulations of the United States Department of Agriculture, or except when said cattle are shipped or transported on railroad trains for immediate slaughter, and when so shipped or transported by railroad, the following regulations must be observed:

On unloading said cattle at their destination, or for food and rest en route to same, separate pens must be set apart to receive them and no other cattle shall be admitted to said pens. The cars that have carried said cattle shall be cleaned and disinfected before they are again used to transport, store or shelter animals or merchandise.

All cars carrying said cattle shall bear placards stating that said cars contain "SOUTHERN CATTLE," and each of the way bills of said shipments shall have a note on its face with a similar statement.

The cars used to transport such cattle shall be cleaned and disinfected in the following manner:

(a) Remove all litter and manure. This litter and manure must be stored where no cattle can come into contact with it for a period of at least eight months.

(b) Wash the cars with water until clean.

(c) Saturate the entire inner surface of the cars, including the inner surface of the car doors, with a mixture made of one and one-half pounds of lime and one-quarter pound of pure carbolic acid to each gallon of water; or with a solution made by dissolving four ounces of chloride of lime to each gallon of water.

It is further ordered that violation of any or either of the foregoing rules and regulations shall be an offense, and punishable as provided by the laws of the State of California.

Now, therefore, I, J. N. Gillett, as Governor of the State of California, by virtue of the authority vested in me by law, do hereby proclaim the foregoing rules and regulations prescribed by the State Veterinarian to be legal and binding rules and regulations within the State of California, and I do further proclaim that said rules and regulations shall be maintained and enforced within the State of California, and that a violation thereof shall subject all persons so violating any of said rules or regulations to the penalties provided for in section eight of that said act of the legislature of the State of California entitled, "An act to protect domestic live stock from contagious and infectious diseases, to provide for the appointment and duties of officials to carry into effect the provisions of this act, and to provide an appropriation therefor."

In witness whereof, I have hereunto set my hand and caused the Great Seal of this State to be hereunto affixed, this first day of March, A. D. 1910.

[Signed.] J. N. GILLETT,
Governor of the State of California.

Attest:
[Signed.] C. F. CURRY,
Secretary of State.

[SEAL.]

STATE OF CALIFORNIA,
OFFICE OF STATE VETERINARIAN,
SACRAMENTO.

PROCLAMATION.

EXECUTIVE DEPARTMENT, STATE OF CALIFORNIA,
SACRAMENTO, CAL., March 1, 1910.

WHEREAS, The State Veterinarian of the State of California has ascertained that cattle located in certain counties and portions of counties in the State of California hereinafter named are liable to communicate an infectious disease known as Texas, splenic or Southern fever, to cattle located in other counties and portions of counties in the State of California should said cattle located in the counties and portions of counties hereinafter named be shipped, moved, transported, driven, or grazed over the lands situated in other counties and portions of counties in the State of California; and

WHEREAS, Under and by virtue of an act of the legislature of the State of California, entitled "An act to protect domestic live stock from contagious and infectious diseases, to provide for the appointment and duties of officials to carry into effect the provisions of this act, and to provide an appropriation therefor," which became a law March 18, 1899, and as amended March 20, 1905, and March 23, 1907, and as

further amended and approved March 19, 1909, the State Veterinarian of the State of California, in order to prevent the spreading or communication of said disease of Texas, splenic or Southern fever, has, on this first day of March, 1910, quarantined the following counties and portions of counties in the State of California:

The entire county of San Diego.

The entire county of Orange.

The entire county of Santa Barbara.

All that portion of San Luis Obispo County lying west and south of a line commencing at the point of the intersection of the township line between township 32 south, range 17 east, and 32 south, range 18 east, and the Santa Maria River on the southern boundary of the county of San Luis Obispo, and extending northerly on said township line between townships 32 south, range 17 east, 32 south, range 18 east, and 31 south, range 17 east, and 31 south, range 18 east, to its intersection with the summit of the Santa Lucia range of mountains; thence following the summit of the Santa Lucia range of mountains northerly and northwesterly to its intersection with the northerly boundary line of San Luis Obispo County.

All that portion of Fresno County situated within township 17 south, range 20 east.

All that portion of Kings County situated in township 20 south, range 22 east.

All that portion of Tulare County situated in townships 21 south, range 27 east and 22 south, range 27 east.

All that portion of San Bernardino County located in the following townships: 9 north, 1 west; 9 north, 1 east; 9 north, 2 east; 10 north, 2 east; 10 north, 3 east; and 10 north, 4 east; and

WHEREAS, The State Veterinarian has, on this first day of March, 1910, made and established the following rules and regulations as to the movements of cattle from said quarantined counties and portions of quarantined counties into other counties or portions of counties in the State of California:

From and after this first day of March, 1910, every person, company, corporation, their agents and servants, are hereby prohibited from driving, leading, moving, carrying or transporting or from causing or permitting to be driven, led, moved, transported, carried or drifted into any other county or portion of county in the State of California, any cattle originating or being in the aforesaid quarantined counties and portions of counties, unless such cattle have first been inspected by the State Veterinarian or his duly authorized deputy and are accompanied by a certificate issued by such officer stating that said cattle are free from contagious and infectious diseases and cattle ticks (*Margaropus annulatus*); provided, however, that when said cattle are shipped or transported on railroad trains for immediate slaughter inspection and certification of such cattle will not be required, but that when so shipped or transported by railroad the following regulations must be observed:

On unloading said cattle at their destination, or for food and rest en route to same, separate pens must be set apart to receive them and no other cattle shall be admitted to said pens. The cars that have carried said cattle shall be cleaned and disinfected before they are again used to transport, store or shelter animals or merchandise.

All cars carrying said cattle shall bear placards stating that said cars contain "SOUTHERN CATTLE," and each of the way bills of said shipments shall have a note on its face with a similar statement.

The cars used to transport such cattle shall be cleaned and disinfected in the following manner:

(a) Remove all litter and manure. This litter and manure must be stored where no cattle can come into contact with it for a period of at least eight months.

(b) Wash the cars with water until clean.

(c) Saturate the entire inner surfaces of the cars, including the inner surface of the car doors, with a mixture made of one and one-half pounds of lime and one quarter pound of pure carbolic acid to each gallon of water; or with a solution made by dissolving four ounces of chloride of lime to each gallon of water.

It is further ordered that a violation of any or either of the foregoing rules and regulations shall be an offense, and punishable as provided by the laws of the State of California.

Now, therefore, I. J. N. Gillett, as Governor of the State of California, by virtue of the authority vested in me by law, do hereby proclaim the counties and portions of counties named in this proclamation quarantined within the State of California, and further hereby proclaim the foregoing rules and regulations prescribed by the

State Veterinarian for the maintenance and enforcement of such quarantine to be legal and binding rules and regulations within the State of California, and I do further proclaim that said rules and regulations shall be maintained and enforced within the State of California, and that a violation thereof shall subject all persons so violating any of said rules or regulations to the penalties provided for in section eight of that said act of the legislature of the State of California entitled, "An act to protect domestic live stock from contagious and infectious diseases, to provide for the appointment and duties of officials to carry into effect the provisions of this act, and to provide an appropriation therefor."

In witness whereof, I have hereunto set my hand and caused the Great Seal of this State to be hereunto affixed, this first day of March, A. D. 1910.

[Signed.] J. N. GILLETT,
Governor of the State of California.

Attest:
[Signed.] C. F. CURRY,
Secretary of State.

[SEAL.]